



Giovanni Gunawan Wangidjaja, S.T.

Mechatronics Engineer | Dual Degree Graduate | Automation & Electrical Systems

PROFILE

Dual-degree Mechatronics Engineering graduate from Swiss German University and Fachhochschule Südwestfalen, with hands-on experience in electrical systems, industrial panel wiring, and manufacturing through internships in both Germany and Indonesia. Proficient in automation, CAD design, and control systems, complemented by practical skills in data analysis and machine learning. Seeking an engineering role in automation, electrical, or manufacturing environments.

WORK EXPERIENCE

Electrical Engineer (Intern)

Mar 2024 – Aug 2024

Schalt und Energietechnik München GmbH (SEM), Germany

- Assembled medium- and low-voltage energy distribution systems, supporting industrial panel installation projects under German safety and quality standards.
- Fabricated small metal components based on technical drawings and engineering specifications.
- Performed final inspection and functional testing during quality control processes.
- Executed drilling, sanding, cutting, and copper bending for electrical system assembly.
- Wired control cabinets, switchgear panels, and energy distribution systems in industrial settings.

MEP Engineer (Intern)

Dec 2022 – Feb 2023

PT Dacrea, Indonesia

- Assisted supervisors in inspecting Mechanical, Electrical, and Plumbing (MEP) systems on construction projects.
- Documented system defects, maintenance issues, and required corrective actions.
- Supported technical coordination to ensure smooth on-site operations and project execution.

Manufacturing & Machining Trainee

Jul 2022 – Aug 2022

Politeknik Industri ATMI, Indonesia

- Operated industrial machines including milling, turning, and benchwork equipment.
- Learned and applied basic welding techniques in a workshop environment.
- Interpreted technical drawings and produced designs using AutoCAD.
- Studied basic CNC machining simulation and manufacturing processes.

EDUCATION

Bachelor of Engineering (B.Eng.), Mechatronics Engineering

2024 – 2026

Fachhochschule Südwestfalen, Germany

Bachelor of Engineering (S.T.), Mechanical Engineering

2021 – 2026

Swiss German University, Indonesia

Senior High School (Science)

2018 – 2021

Canisius College, Indonesia

TECHNICAL SKILLS

CAD & Design:

AutoCAD, SolidWorks, FreeCAD, PCB Design (Proteus), Technical Drawing

Engineering:

PLC, Arduino IDE, ESP32, Machining (Milling, Turning, Benchwork)

Data & Programming:

Python, SQL, Data Visualization, Statistical Analysis, Machine Learning

Tools:

Microsoft Excel, Git & GitHub, Visual Studio Code

CERTIFICATIONS

Siemens Mechatronic Systems Certification Program (SMSCP) – Level 1

2024

Siemens, Germany

Data Science & Machine Learning Certification

2025

Purwadhika Digital Technology School, Indonesia

PROJECTS

Self-Balancing Robot – Thesis Project

2026

3D modeled and printed robot chassis (SolidWorks, FreeCAD), wired ESP32 microcontroller, and implemented two control systems (PID vs Fuzzy Logic) with Python Teleplot for data extraction and analysis.

Bank Marketing Campaign — Data Science Project

2025

Built a machine learning model to predict term deposit subscription using Logistic Regression, KNN, Decision Tree, and Random Forest.

LANGUAGES

English — Fluent

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Indonesian — Native

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German — Basic

LINK

- **GitHub:** <https://github.com/Giovanni-ivanong>
- **LinkedIn:** www.linkedin.com/in/giovanni-gunawan-wangidjaja-145060352
- **Portfolio:** <https://giovanni-ivanong.github.io>